

Taiyo Williamson

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EDUCATION

University of San Francisco, San Francisco, CA

B.S. in Computer Science, GPA: 3.69

M.S. in Computer Science, CS 4+1, GPA: 3.67

Aug 2022 - Present

Graduated May 2026

Expected graduation May 2027

Relevant Coursework:

- Undergraduate: C & Systems Programming, Data Structures & Algorithms, Full-Stack Web Dev, Comp Architecture
- Graduate: AI Foundations, ML Foundations, AI Redistricting, Algorithms, Computer Networks, Systems Foundations

EXPERIENCE

CS Department, University of San Francisco, CA | *VGL Research Assistant*

Feb 2026 - Present

- Architecting cloud tooling for USF lab infrastructure and leading development of Nala, an AI health coaching chatbot.
- Coauthored a publication on Nala at ACM SIGCHI, and presented at the Dean's Reception for Donors event.
- Pursuing a Master's Thesis on Nala's cross-cultural applications and system optimization.

CS Department, University of San Francisco, CA | *CS Tutor & TA*

Jan 2025 - May 2026

- Supported and provided tutoring in debugging, project management and development, and exam prep.
- Hosted workshops on SSH & Git basics, developed lesson plans and coding exercises to reinforce core concepts.

Hack4Impact, University of San Francisco, CA | *Chapter Founding Executive Director*

Jan 2024 - May 2026

- Founded the university's 3rd computer science club; mimics a professional working environment to provide exposure.
- Taught students a React-Next.js-AWS-oriented tech stack to develop software solutions for local nonprofits.

Fastly, San Francisco, CA | *Embedded Software Engineering Intern*

May 2025 - Aug 2025

- Collaborated with the Embedded Fastly team to optimize, scale, and test virtual Fastly's global CDN servers.
- Designed and built a Go-based API cache testing tool from the ground up, integrating directly with cache nodes across deployment types to test compute and observability stack components.
- Deployed the tool company-wide via Fastly's CI/CD pipeline, hosting it as an internal API on a private Kubernetes cluster with a custom Helm chart for streamlined provisioning on new machines.

Pacxa, Honolulu, HI | *Software Developer Intern* | [App Demo](#)

June 2024 - Aug 2024

- Leveraged Microsoft Power Apps to develop a business-purpose talent and certification management tool.
- Developed Power Automate flows to automate email, Excel, and Office 365 tasks, decreasing manual labor by 80%.
- Implemented RBAC for Pacxa's 50+ employees, eliminating unauthorized access.

PROJECTS

Nala: [Site](#) | [Github](#)

Aug 2025

- Built a full-stack AI health coaching app using Claude Sonnet 4.5 with a RAG pipeline backed by 58 real health coach transcripts, enabling contextually grounded, session-aware conversations across a 4-session coaching program.
- Created a FastAPI-based RESTful API layer that navigates between the React frontend and the RAG system backend.
- Architected the AI backend with pgvector-powered semantic search, session state machines, SMART goal detection/refinement, and conversation persistence.

Network Compression Detector: [Server-Client](#) | [Standalone](#)

May 2024

- Created 2 applications using C Socket Programming to detect compression links on network paths with 100% accuracy.
- Using UTM VMs with Ubuntu Linux, the server VM initiates a TCP connection and sends a high and low entropy UDP payload train for the client VM to receive all the packets. Wireshark captures the packet transactions.
- Replicated and tested compression detector code in Golang for self-learning, and ran performance tests for efficiency.

TECHNICAL SKILLS

- Programming Languages: Golang, C, Python, Java, Rust, RISC-V, React JavaScript, R, HTML, CSS
- Software/Framework: Kubernetes, Helm, Jenkins, Terraform, Docker, Node.js, Git, GCP, AWS, Supabase, Ubuntu
- Languages: Proficient in Japanese